The Fine Structure Constant is Not a Constant: Ghost Redistribution and the Webb Dipole

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Abstract

We show that the observed spatial variation of the fine structure constant α_{em} (the Webb dipole, $\Delta\alpha/\alpha \sim 10^{-5}$) is a natural consequence of the DRLT ghost redistribution mechanism. The fundamental coupling $\alpha_{\text{GUT}} = 6/(25\pi^2)$ is a true constant, protected by trace conservation ($\text{Tr}(G) = N$). However, the individual couplings $\alpha_3, \alpha_2, \alpha_1$ receive ghost corrections δ_i that sum to zero but whose *spatial distribution* can vary. A variation $\Delta\varepsilon_0/\varepsilon_0 \sim 0.17\%$ — comparable to the CMB dipole ($v/c = 0.12\%$) — reproduces the observed $\Delta\alpha/\alpha \sim 10^{-5}$. The theory makes a falsifiable prediction: α_s must vary in the *opposite direction* to α_{em}, maintaining $\sum \delta_i = 0$ at every spatial location. This can be tested with existing quasar absorption line data.

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1 Introduction: Feynman's Question

Richard Feynman called the fine structure constant $\alpha_{\mathrm{em}} \approx 1/137.036$ a "magic number" that all physicists should lose sleep over. For 70 years, the question "why 1/137?" has had no answer. The Standard Model treats it as an input parameter — a number that must be measured, not derived.

In 1999, Webb et al. reported evidence that α_{em} is not even constant: quasar absorption spectra suggest a spatial variation $\Delta\alpha/\alpha \sim 10^{-5}$ with a dipole pattern across the sky [1]. This finding, if confirmed, would overturn the assumption that physical laws are identical everywhere. For 25 years, the community has been split: is the Webb dipole real physics or systematic error?

We propose a resolution from Dynamic Resolution Lattice Theory (DRLT) [2]: **α_{em} is not a fundamental constant. It is a local manifestation of ghost redistribution.** The true constant is $\alpha_{\mathrm{GUT}} = 6/(25\pi^2)$, derived from lattice geometry. The measured α_{em} varies because the ghost corrections δ_i — remnants of annihilated $N > 5$ structure — are spatially non-uniform. Both sides of the Webb debate are correct: α_{em} varies (Webb is right), but the underlying physics is universal (α_{GUT} is constant).

2 The Ghost Sum Rule (Review)

2.1 Coupling constants from DRLT

In DRLT [2], the three gauge couplings at M_Z are derived from the Binet–Cauchy decomposition of $\Lambda^3(\mathbb{C}^5)$:

$$1/\alpha_3 = 1 \times 8 \times S(1) = 8.0, \tag{1}$$
$$1/\alpha_2 = 12 \times 2 \times S(2) = 30.0, \tag{2}$$
$$1/\alpha_1 = 12 \times 3 \times S(\infty) = 6\pi^2 \approx 59.2, \tag{3}$$

where $S(N) = \sum_{n=1}^{N} 1/n^2$ is the partial Basel sum.

These "plateau values" differ from observation by the ghost corrections:

$$\delta_3 = (1/\alpha_3)_{\mathrm{obs}} - 8.0 = +0.47, \tag{4}$$
$$\delta_2 = (1/\alpha_2)_{\mathrm{obs}} - 30.0 = -0.40, \tag{5}$$
$$\delta_1 = (1/\alpha_1)_{\mathrm{obs}} - 59.2 = -0.22. \tag{6}$$

2.2 The sum rule

The ghost corrections obey a sum rule:

$$\sum_i \delta_i = \delta_3 + \delta_2 + \delta_1 + \delta_G = +0.47 - 0.40 - 0.22 + 0.15 = 0.00, \tag{7}$$

proven from trace conservation: $\mathrm{Tr}(G) = N$ is invariant under the unitary projection from rank N to rank 5.

2.3 The base perturbation ε_0

All ghost corrections scale with a single parameter $\varepsilon_0 \approx 0.0038$:

$$\delta_i = \mathrm{Sgn}_i \times (1/\alpha_i)_{\mathrm{pred}} \times M_i \times \varepsilon_0, \tag{8}$$

where M_i are geometric weights determined by $(n_S, n_T) = (3, 2)$. The parameter ε_0 is not free: it encodes the local "diagonalization progress" of the Gram matrix — the cosmic address.

3 Spatial Variation of Ghosts

3.1 Key insight: ε_0 is a local field

The ghost parameter ε_0 measures how far the local Gram matrix has progressed in its diagonalization (from rank N toward rank 1). In the block universe, different spatial regions can be at slightly different stages:

$$\varepsilon_0 = \varepsilon_0(\mathbf{x}), \tag{9}$$

varying with position due to the large-scale distribution of matter (which affects the local $\det(G_h)$ structure).

3.2 What varies and what doesn't

Theorem 3.1 (Spatial Ghost Sum Rule). *At every spatial position* $\mathbf{x}$*:*

$$\sum_i \delta_i(\mathbf{x}) = 0. \tag{10}$$

The total coupling $1/\alpha_{GUT} = 25\pi^2/6$ is spatially invariant. Only the distribution *of ghosts among sectors varies.*

Proof. $\mathrm{Tr}(G) = N$ is a global constraint, independent of position. The Binet–Cauchy completeness $\sum_{\text{channels}} = d^2 = 25$ is a geometric identity, independent of the state ψ. Therefore $\sum_i (1/\alpha_i) = 25\pi^2/6$ at every point, and $\sum_i \delta_i = 0$ everywhere. □

Corollary 3.2. *The fine structure constant*

$$1/\alpha_{em} = \tfrac{5}{3}(1/\alpha_1) + (1/\alpha_2) = \tfrac{5}{3}(59.2 + \delta_1) + (30.0 + \delta_2) \tag{11}$$

varies spatially because $\delta_1(\mathbf{x})$ and $\delta_2(\mathbf{x})$ vary, even though their contribution to α_{GUT} is fixed.

4 Quantitative Comparison with the Webb Dipole

4.1 Observed signal

Webb et al. report a dipole variation in α_{em} across the sky:

$$\frac{\Delta\alpha}{\alpha} \approx (1.1 \pm 0.25) \times 10^{-5}, \tag{12}$$

with the dipole axis pointing toward $(\mathrm{RA}, \mathrm{Dec}) \approx (17.3\mathrm{h}, -61^\circ)$ [1].

4.2 DRLT prediction

If ε_0 varies by a fraction $f = \Delta\varepsilon_0/\varepsilon_0$ across the sky:

$$\Delta\delta_1 = \delta_1 \times f = -0.22\, f, \tag{13}$$

$$\Delta\delta_2 = \delta_2 \times f = -0.40\, f, \tag{14}$$

$$\Delta(1/\alpha_{\mathrm{em}}) = \tfrac{5}{3}\,\Delta\delta_1 + \Delta\delta_2 = -0.767\, f. \tag{15}$$

The fractional variation:

$$\frac{\Delta\alpha_{\mathrm{em}}}{\alpha_{\mathrm{em}}} = \frac{0.767\, f}{128.7} = 5.96 \times 10^{-3}\, f. \tag{16}$$

Setting this equal to the observed 10^{-5}:

$$f = \frac{10^{-5}}{5.96 \times 10^{-3}} = 1.68 \times 10^{-3} = 0.17\%. \tag{17}$$

4.3 Comparison with the CMB dipole

The CMB dipole (our velocity relative to the CMB rest frame) gives:

$$\frac{v}{c} = \frac{370\ \mathrm{km/s}}{3 \times 10^5\ \mathrm{km/s}} = 1.23 \times 10^{-3} = 0.12\%. \tag{18}$$

The required ε_0 variation (0.17%) is **1.4 times the CMB dipole** — the same order of magnitude. This is non-trivial: the ghost redistribution parameter varies at a scale consistent with the known large-scale anisotropy of the universe.

4.4 Direction

The Webb dipole axis $(17.3\mathrm{h}, -61^\circ)$ does not align with the CMB dipole $(11.2\mathrm{h}, -7^\circ; \sim 100^\circ$ separation). In DRLT, this is expected: ε_0 depends on the local $\det(G_h)$ distribution (large-scale structure), which has its own dipole independent of our peculiar velocity.

5 Falsifiable Predictions

The ghost redistribution mechanism makes four testable predictions that distinguish it from all other models of varying constants:

5.1 Prediction 1: α_s anti-correlates with $\alpha_{\mathbf{em}}$

The spatial ghost sum rule $\sum_i \delta_i(\mathbf{x}) = 0$ requires that if α_{em} is larger in some direction, α_s must be **smaller** in that direction:

$$\Delta\delta_3 = -(\Delta\delta_1 + \Delta\delta_2 + \Delta\delta_G) \approx -\Delta\delta_1 - \Delta\delta_2. \tag{19}$$

This is testable: molecular hydrogen lines in the same quasar spectra can probe α_s (through the proton-to-electron mass ratio $\mu = m_p/m_e$, which depends on Λ_{QCD}, which depends on α_s).

If $\Delta\alpha_{\mathbf{em}}$ and $\Delta\mu$ have the same sign, DRLT is falsified.

5.2 Prediction 2: $\alpha_{\mathbf{GUT}}$ is spatially invariant

The combination

$$\frac{1}{\alpha_{\text{GUT}}} = \frac{1}{\alpha_3} + \frac{12 \times 2}{1 \times 8}\frac{1}{\alpha_2} + \frac{12 \times 3}{1 \times 8}\frac{1}{\alpha_1} = \frac{25\pi^2}{6} \tag{20}$$

must be identical at every spatial location. Any observed variation in this combination would falsify the trace conservation mechanism.

5.3 Prediction 3: Dipole amplitude depends on redshift

The ghost parameter $\varepsilon_0(z)$ changes with cosmic epoch (redshift z). The dipole amplitude should therefore vary with z:

$$\frac{\Delta\alpha}{\alpha}(z) = 5.96 \times 10^{-3} \times \frac{\Delta\varepsilon_0(z)}{\varepsilon_0(z)}. \tag{21}$$

At high z (early universe, larger ε_0), the fractional variation $\Delta\varepsilon_0/\varepsilon_0$ could differ, producing a z-dependent dipole. This is testable with high-z quasar surveys.

5.4 Prediction 4: No variation in $\alpha_{\mathbf{em}}$ at the GUT scale

At energies approaching M_{GUT}, all forces see $N_{\text{eff}} = \infty$, ghosts are uniformly distributed, and $\varepsilon_0 \to 0$. Therefore $\Delta\alpha/\alpha \to 0$ at high energy. If constant variation is observed in early-universe processes (e.g., BBN), DRLT is falsified.

6 The True Constant

Feynman asked: "Why 1/137?" DRLT answers: **the question is wrong.**

1/137 is not a fundamental constant. It is a local value of $1/\alpha_{\text{em}}$, determined by the ghost redistribution at our cosmic address ($\varepsilon_0 \approx 0.0038$). The fundamental constant is:

$$\boxed{\frac{1}{\alpha_{\text{GUT}}} = \frac{25\pi^2}{6} = d^2 \times \zeta(2) \approx 41.12} \tag{22}$$

derived from (i) $d = 5$ (the unique dimension, proven from atomic number theory), (ii) $\zeta(2) = \pi^2/6$ (Euler's Basel sum, the total lattice illuminance per channel).

Remark 6.1 (A constant of the plateau, not the universe). Even $25\pi^2/6$ is not a universal constant. It is a geometric identity of the rank-5 plateau — the long-lived intermediate state that survives swap annihilation of repeated atomic dimensions. The universe itself is rank N (arbitrarily large); the rank-5 plateau is a transient structure, albeit one that persists long enough for chemistry, biology, and observers. In the block universe, regions of rank $\gg 5$ (Big Bang) and rank $\to 1$ (heat death) coexist with the plateau. The "constant" $25\pi^2/6$ holds wherever and whenever the rank-5 structure is active — which, for all practical purposes in our epoch, is everywhere we can observe.

This number is:

- Derived, not measured.
- Universal, not local.
- Protected by trace conservation.
- Spatially invariant (by theorem).

The hierarchy of constants:

Quantity	Status	Why
(none)	**No universal constant**	Universe = rank-N matrix
$25\pi^2/6$	**Plateau constant**	Geometric identity of rank-5
$1/\alpha_3, 1/\alpha_2, 1/\alpha_1$	Local values	Ghost redistribution
$1/\alpha_{\text{em}} \approx 137$	Local composite	$= \frac{5}{3}(1/\alpha_1) + (1/\alpha_2)$
1/137.036	Our address	$\varepsilon_0 \approx 0.0038$ here, now

The rank-5 plateau is itself a consequence of swap annihilation: all atomic dimensions except one copy of 2 and one copy of 3 are destroyed by symmetry. The surviving $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ structure is slowly decaying: SU(3) is already confined ($N_{\text{eff}} = 1$), SU(2) is broken ($N_{\text{eff}} = 2$, W/Z massive), and U(1) will eventually lose its phase coherence in the far future. The plateau is not eternal — it is the era of maximum complexity between the rank-N chaos and the rank-1 vacuum.

7 Conclusion

The Webb dipole is not a systematic error. It is not evidence for new fields or extra dimensions. It is the spatial non-uniformity of ghost redistribution in $\mathbb{C}^5$, visible because α_{em} is a composite of ghost-corrected couplings.

The 25-year debate between "α varies" and "α is constant" is resolved: both are right, about different quantities. α_{em} varies. α_{GUT} is constant. The difference is ghosts.

The theory makes an immediately testable prediction: α_s must anti-correlate with α_{em} spatially, preserving $\sum \delta_i = 0$. This can be checked with existing quasar data. A positive result would confirm the ghost mechanism; a negative result would falsify it.

The fine structure constant is not a constant.
It is a ghost's shadow, cast differently in different places.
$25\pi^2/6$ *is the constant of our plateau — the rank-5 era.*
The universe itself has no constants. Only structure.

References

[1] J. K. Webb et al., "Indications of a Spatial Variation of the Fine Structure Constant," Phys. Rev. Lett. **107**, 191101 (2011).

[2] M. Jeong, "A Note on Lattice Geometry in $\mathbb{C}^5$" (2026).